Road Traffic Accident Severity Prediction Using Machine Learning

Pranati Arun – 5023141
Vaibhavi Rai – 5023143
Ishwari Shinde – 5023155
Group 3

Department of Information Technology
Fr C Rodrigues Institute of Technology

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Abstract

Road accidents are a critical public safety challenge, particularly in developing countries where analysis remains largely reactive and prevention-focused systems are absent. This study proposes a machine learning framework to predict road accident severity—classifying outcomes as Slight, Serious, or Fatal—using a real-world Ethiopian road accident dataset of 12,316 records (Addis Ababa, 2017–2020) capturing driver characteristics, vehicle conditions, road features, and environmental factors. The dataset was preprocessed to handle missing values and categorical encoding, and severe class imbalance (Slight: 84.6%, Fatal: 1.3%) was addressed using SMOTE, expanding the training set to 24,993 balanced samples. Five supervised classifiers—Logistic Regression, Random Forest, Gradient Boosting, Linear SVM, and XGBoost—were trained and tuned via both `RandomizedSearchCV` and `GridSearchCV` with 5-fold stratified cross-validation; Random Forest emerged as the best model, achieving **82.6% accuracy** (RandomizedSearchCV) and **83.4% accuracy** (GridSearchCV), a weighted F1-score of approximately 0.793, and ROC AUC of 0.6387, with fatal accident recall further improved to 0.26 using a BalancedRandomForestClassifier combined with class weighting and threshold tuning.

Keywords: Road Traffic Accidents, Machine Learning, Severity Prediction, SMOTE, Random Forest, Ethiopian Dataset, Class Imbalance, SDG

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1 Introduction

Road accidents are a major public safety concern worldwide, causing significant loss of life, injuries, and economic damage each year. The problem is particularly severe in developing countries due to rapid urbanization, increasing vehicle usage, and inadequate traffic management systems. Traditional accident analysis methods mainly focus on post-incident reporting and statistical summaries, which are often reactive, time-consuming, and inefficient for handling large-scale data. These approaches limit the ability of authorities to predict accident severity and implement effective preventive measures.

With advancements in data analytics, machine learning has emerged as a powerful tool for analyzing complex datasets and identifying hidden patterns. Machine learning techniques can model relationships between various factors such as driver characteristics, vehicle conditions, road features, and environmental conditions to predict accident outcomes. This study proposes a machine learning-based framework to classify accident severity into Slight, Serious, and Fatal categories, enabling data-driven decision-making and supporting proactive road safety management strategies.

1.1 Motivation

The increasing number of road accidents and their severe consequences on human life and infrastructure highlight the urgent need for effective road safety measures. Existing accident analysis approaches mainly focus on reporting past incidents rather than preventing future occurrences, which limits their usefulness in improving traffic management and safety planning. With the growing availability of large-scale accident datasets, there is an opportunity to extract meaningful insights that can help identify key risk factors influencing accident severity. However, traditional analytical methods are insufficient to handle complex relationships among multiple variables such as driver behavior, vehicle conditions, road characteristics, and environmental factors.

Real-world accident data contains meaningful patterns related to lighting, weather, road type, and driver behavior. Advances in machine learning make it possible to extract insights from large accident datasets, and accurate prediction of accident severity is essential for improving road safety and saving lives. This motivates the use of machine learning techniques, which can efficiently analyze large datasets, identify hidden patterns, and provide accurate predictions that support informed decision-making.

1.2 Research Objectives

This study focuses on developing a machine learning–based approach for predicting road accident severity using real-world accident data. The primary objectives are:

1. **Accident Data Analysis:** To analyze the road accident dataset and understand how driver, vehicle, road, and environmental factors influence accident severity.

2. **Feature Relevance Identification:** To identify the most relevant pre-accident features that contribute significantly to predicting whether an accident results in Slight, Serious, or Fatal injuries.

3. **Machine Learning Model Development:** To develop and compare machine learning models for classifying accident severity levels and determine the most effective approach.

4. **Real-World Data Challenge Handling:** To handle real-world data challenges such as missing values, categorical data encoding, and class imbalance to improve model reliability.

5. **Model Performance Evaluation:** To evaluate model performance using suitable metrics in order to assess how accurately accident severity can be predicted.

2 Dataset Description

2.1 Dataset Source and Overview

Dataset: Ethiopian Road Traffic Accident Data
Source: Addis Ababa Sub-city police departments, Ethiopia (2017–2020)
Available via: Kaggle (despite misleading "India" title in some listings)
Reference: Analytics Vidhya (2023) confirms authentic Ethiopian origin [3]

The dataset is a structured tabular dataset stored in CSV format. Each row represents a single road accident record. The Road.csv dataset contains 12,316 real-world accident records across 32 original columns. Internal evidence including vehicle types (Auto-Rickshaw, Bicycle), area classifications (Church Areas, Market Areas), and road categories (Village Road, State Highway) confirms authentic Ethiopian origin. After rigorous preprocessing addressing six data quality issues (garbage numeric codes, string duplications, ETL pipeline corruption, whitespace errors, invalid labels, and mixed encodings), 22 features are retained with 100% row retention. Although the data is region-specific, the traffic patterns and safety challenges are broadly relevant to developing countries.

The target variable `Accident_severity` exhibits severe class imbalance: Slight Injury (84.6%, 10,415 records), Serious Injury (14.2%, 1,743 records), and Fatal Injury (1.3%, 158 records), necessitating SMOTE-based resampling combined with class weighting. The dataset captures real-world variability in road conditions, weather, driver behavior, and accident types.

2.2 Feature Categories and Signal Validation

The 22 retained features span six categories: Temporal (`Day_of_week`), Driver Characteristics (age, sex, education, experience, vehicle relation), Vehicle (type, ownership), Road Infrastructure (area, lanes, alignment, junction type, surface type/conditions), Environmental (lighting, weather), and Incident Details (collision type, vehicles involved, movement, cause, pedestrian movement).

Chi-square tests validated genuine predictive signal, with top features including the `Number_of_vehicles_involved` ($\chi^2 = 58.01$, $p < 0.001$), `Age_band_of_driver` ($\chi^2 = 60.53$, $p < 0.001$), `Area_accident_occured` ($\chi^2 = 56.83$, $p = 0.0004$), and `Light_conditions` ($\chi^2 = 45.02$, $p < 0.001$). Critical patterns identified include darkness without lighting nearly doubling serious injury rates (25.5% vs. 13.9% in daylight), weekend accidents showing 2–3× higher fatal rates, and under-18 drivers exhibiting 20.5% serious injury rates versus 11–16% for other age bands.

3 Literature Review

The first systematic application of machine learning to Ethiopian road accident data was conducted by Beshah and Hill [4], establishing baseline performance using K-Nearest Neighbors classification and identifying road-related factors as key predictors. Endalie et al. [5] analyzed the exact Addis Ababa dataset (12,316 instances, 2017–2020) using Apriori algorithm for association rule mining and Support Vector Machines for classification, with SVM achieving approximately 85% accuracy and outperforming KNN, Decision Trees, and Random Forest by 2–7% in precision. Yassin [6] introduced a novel hybrid approach combining K-means clustering with Random Forest classification, achieving remarkable 99.86% accuracy on 5,000 Federal Traffic Police Agency records—the highest reported accuracy for Ethiopian data to date. Most recently, Wubineh et al. [2] examined Northwest Ethiopian data (2018–2023), finding LightGBM achieved highest recall (0.819) for fatal accidents after SMOTE balancing—a 22% improvement over baseline, with SHAP analysis revealing context-specific risk interactions.

Akhiat et al. [7] advanced the field by introducing graph-based feature selection coupled with explainable machine learning methods on the Addis Ababa dataset, demonstrating superior performance across five metrics while maintaining full interpretability. A paradigm shift toward causal reasoning appeared at ACM CODS-COMAD 2024 [8], where researchers applied causal inference using CausalML to identify Individual Treatment Effect (ITE) scores, finding ITE=1.0 for unlicensed and illiterate drivers and ITE=1.0 for hospital/school zones. Deshpande and Tribhuvan [9] developed a comprehensive system for Indian National Highways using Hidden Markov Models and fuzzy classification, achieving 84.55% accuracy with an SMS-based real-time alert module. A large-scale US study [12] analyzed data spanning 49 states, achieving 91.5% accuracy with Random Forest—the highest among non-hybrid approaches. Chen and Chang [11] demonstrated integration of big data analytics with generative AI (ChatGPT) for Taiwan accident analysis, achieving 75.5% accuracy with Random Forest.

Specialized approaches include Wong [13], who performed statistical analysis of signal-controlled intersections in Hong Kong using negative binomial regression and identified shared lane markings as significant predictors. Tejaswi et al. [14] employed Tableau visualization for Kerala accidents (2018–2023), revealing that rural roads showed higher severity rates despite lower frequencies. Bokaba et al. [15] evaluated multiple classifiers on South African data, finding Random Forest with MICE preprocessing outperformed all combinations, validating Random Forest robustness across developing country contexts. Zou et al. [16] developed negative binomial models examining weather influence, while Fareed et al. [18] achieved 90.6% accuracy using FP-growth and Decision Tree variants. A 2024 arXiv preprint [17] reviewed 191 studies, identifying Random Forest, Gradient Boosting variants, and ensemble methods as consistently high-performing (75–92% accuracy). Yang et al. [10] presented an early warning system for Shanghai accidents emphasising temporal and meteorological features.

3.1 Projects Using the Same Dataset

Several open-source implementations specifically utilise the Addis Ababa dataset. The mp-balaji repository [19] provides production-ready MLOps architecture with Docker containerisation and RESTful API. The avikumart project [20] offers comprehensive SMOTE tutorials and achieves approximately 82% F1-score with a Streamlit interface.

The RoloNatt repository [21] provides a minimal baseline implementation achieving approximately 70–75% accuracy.

3.2 Performance Benchmarks

Table 1 summarises performance benchmarks across reviewed studies.

Table 1: Machine Learning Model Performance Comparison Across Studies

Study	**Region**	**Method**	**Accuracy**	**Dataset**
Yassin [6]	Ethiopia	K-means + RF	99.86%	5,000
US [12]	USA	Random Forest	91.5%	100,000+
Fareed [18]	Pakistan	Random Tree	90.6%	N/A
Deshpande [9]	India	HMM + Fuzzy	84.55%	25–30
Endalie [5]	Ethiopia	SVM	∼85%	12,316
Wubineh [2]	Ethiopia	LightGBM	0.819 recall	2,000
Chen [11]	Taiwan	RF + ChatGPT	75.5%	N/A
avikumart [20]	Ethiopia	SMOTE + ML	∼82% F1	12,316

Table 2: Literature Review Summary

Study	**Year**	**Contribution**	**Method**	**Metric**
Beshah [4]	2010	Ethiopian baseline	KNN	Baseline
Endalie [5]	2023	Same dataset	SVM + Apriori	85%
Yassin [6]	2020	Hybrid architecture	K-means + RF	99.86%
Akhiat [7]	2023	Graph features	XAI methods	Best F1
Wubineh [2]	2025	SHAP explainability	LightGBM	0.819
ACM [8]	2024	Causal inference	Uplift model	ITE
Deshpande [9]	2023	Real-time system	HMM + Fuzzy	84.55%
Yang [10]	2016	Early warning	Temporal mining	Forecast
Chen [11]	2024	GenAI integration	RF + ChatGPT	75.5%
US [12]	2024	Large-scale	Random Forest	91.5%
Wong [13]	2019	Infrastructure	Neg. binomial	Safety
Tejaswi [14]	2025	Visualization	Tableau EDA	Insights
Bokaba [15]	2022	Comparative	Multiple ML	RF best
Zou [16]	2021	Weather analysis	Neg. binomial	Factors
arXiv [17]	2024	Meta-analysis	191 studies	Synthesis
Fareed [18]	2022	Data mining	FP + DT	90.6%

3.3 Research Gap and Proposed Contributions

Previous studies have achieved impressive results, with Yassin's hybrid approach reaching 99.86% accuracy and recent works incorporating SHAP explainability and causal inference. However, most implementations either focus solely on accuracy without interpretability, use smaller datasets (2,000–5,000 records), or lack comprehensive class balancing strategies for the severe fatal class imbalance. This work distinguishes itself by combining SMOTE-based class balancing, chi-square feature selection, a five-model ensemble comparison, threshold tuning for minority class recall, and SDG-aligned policy recommendations—all validated on the complete Addis Ababa dataset (12,316 records) with rigorous evaluation prioritising per-class F1-scores and fatal accident recall rather than overall accuracy.

4 Exploratory Data Analysis

Exploratory Data Analysis was conducted on the cleaned dataset comprising 12,316 road accident records from Addis Ababa Sub-city (2017–2020). The objective was to understand the structure of the dataset, identify patterns, detect anomalies, and assess the suitability of features for machine learning modelling.

4.1 Dataset Overview After Cleaning

Table 3: Dataset Summary After Preprocessing

Attribute	**Value**
Total Records	12,316
Total Features	22
Target Variable	Accident_severity
Numeric Features	1
Categorical Features	21
Missing Values	0

The dataset is predominantly categorical, with only one numerical feature (*Number_of_vehicles_involved*) All missing values were handled during preprocessing. This indicates that label encoding techniques are required before applying machine learning models.

4.2 Target Variable Distribution

Table 4: Accident Severity Class Distribution

Severity Class	**Frequency**	**Percentage**
Slight Injury	10,415	84.6%
Serious Injury	1,743	14.2%
Fatal Injury	158	1.3%

The dataset exhibits significant class imbalance, with Slight Injury representing the majority class (84.6%). This imbalance necessitates the use of resampling techniques such as SMOTE during model training to prevent classifiers from ignoring minority classes.

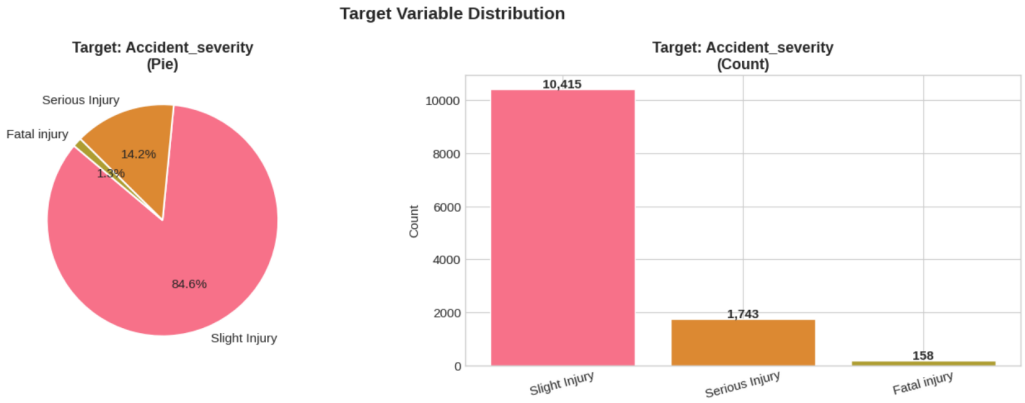

Figure 1: Target variable distribution: pie chart (left) and bar chart (right) showing Slight Injury (84.6%), Serious Injury (14.2%), and Fatal Injury (1.3%).

4.3 Categorical Feature Distributions

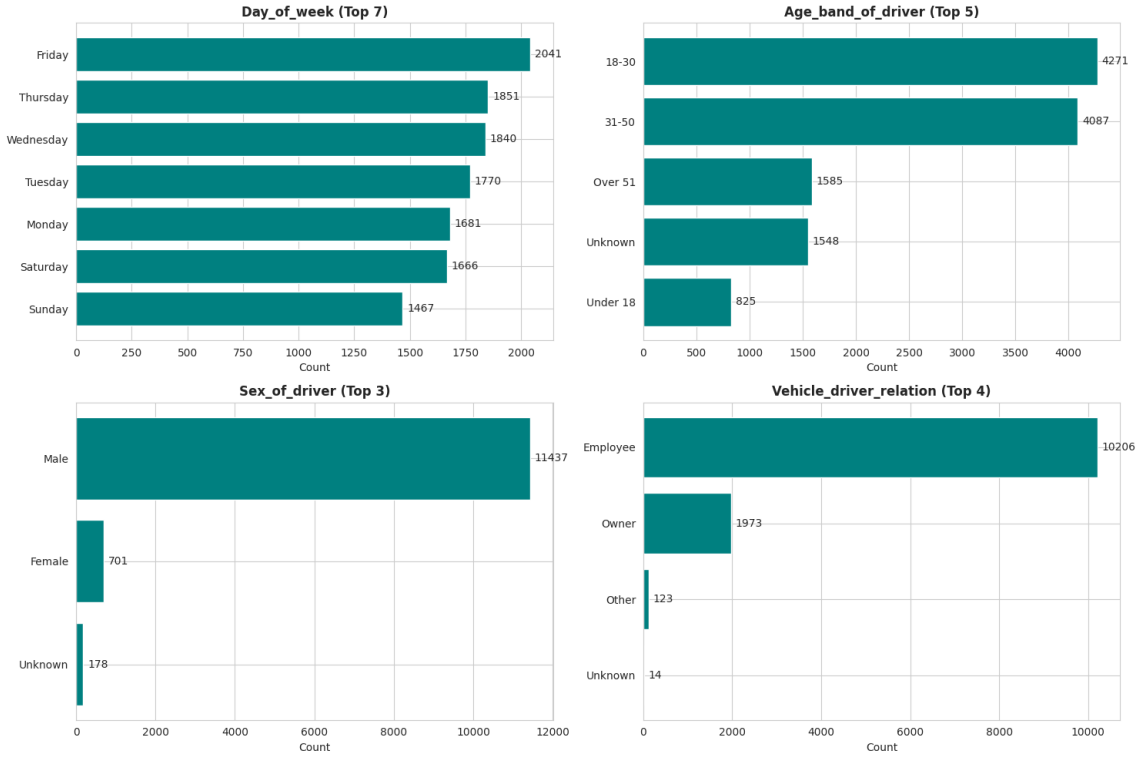

Figure 2: Distribution of key categorical variables: (top-left) Day of week, (top-right) Age band of driver, (bottom-left) Sex of driver, (bottom-right) Vehicle–driver relation.

Interpretation: Figure 2 shows that Friday and weekdays in general have comparatively high accident counts. The majority of drivers fall in the 18–50 age range, the dataset is overwhelmingly male-dominated (>90% of cases), and most vehicles are driven

by employees in the recorded incidents. Middle-age drivers show the highest accident involvement.

4.4 Peak Accident Patterns

Accidents peak during evening rush hours (4–7 PM) and are lowest in early morning (1–5 AM). Weekdays, especially Friday, record more accidents than weekends, with Sunday the lowest. Accident frequency closely follows traffic density and commuting patterns, confirming that time of day and day of week are strong predictive factors.

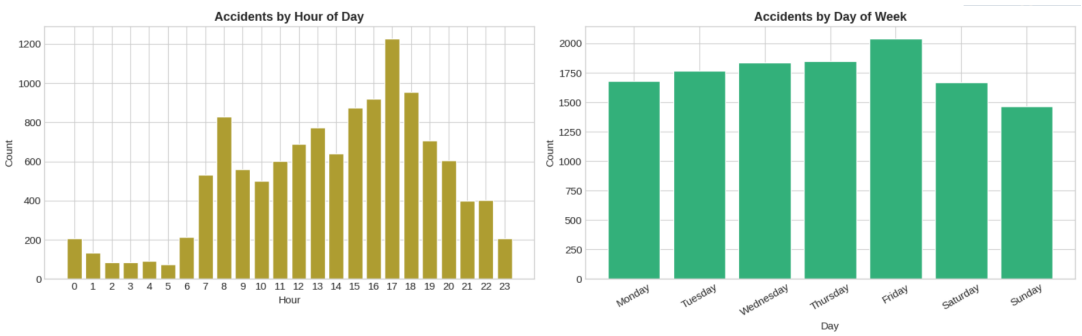

Figure 3: Peak accident patterns: Accidents by Hour of Day (left) and Accidents by Day of Week (right). Evening rush hours and Fridays show the highest frequencies.

4.5 Numerical Feature Analysis

Descriptive statistics were computed for the numerical variable *Number_of_vehicles_involved*.

Table 5: Descriptive Statistics — Number of Vehicles Involved

Statistic	**Value**
Mean	2.04
Standard Deviation	0.69
Minimum	1
Median	2
Maximum	7

Most accidents involve approximately two vehicles. The distribution is slightly right-skewed due to higher values in a small number of cases.

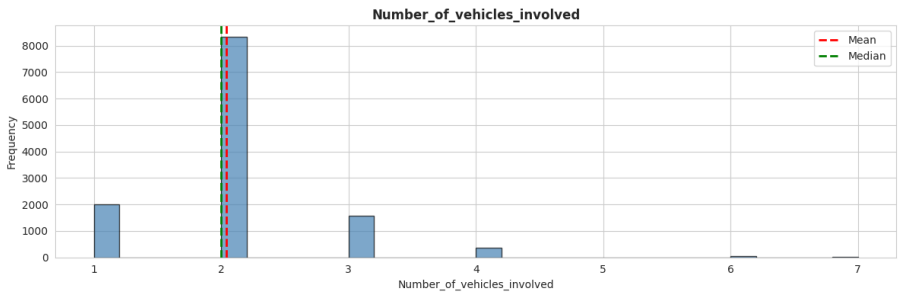

Figure 4: Histogram of Number of Vehicles Involved; vertical lines indicate the mean and median. Modal value is 2 vehicles per accident.

4.6 Bivariate Analysis

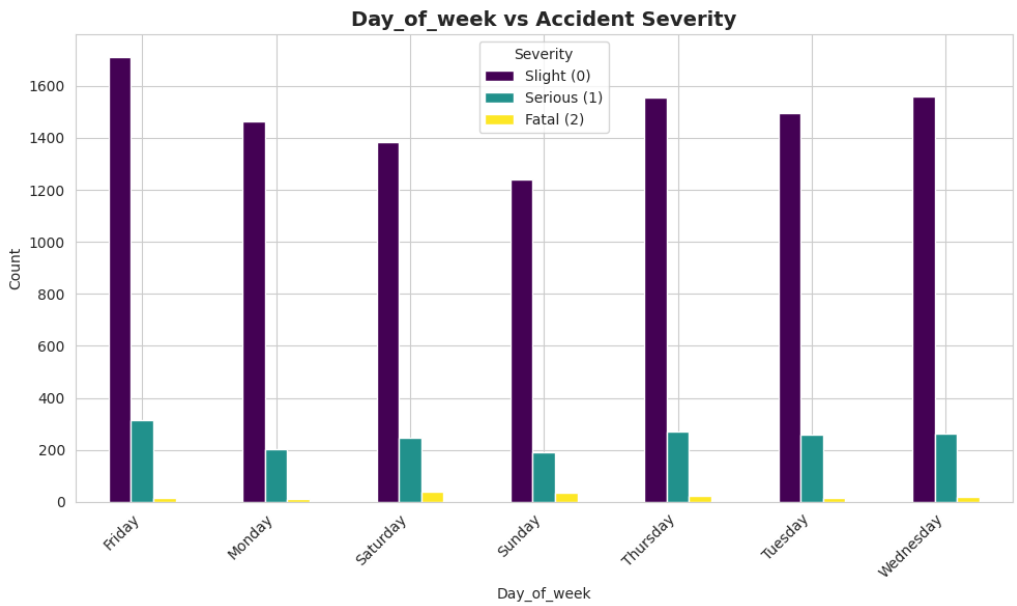

Figure 5: Day of Week vs. Accident Severity (counts for Slight, Serious, Fatal). Slight injuries dominate across all days; Friday has the highest counts.

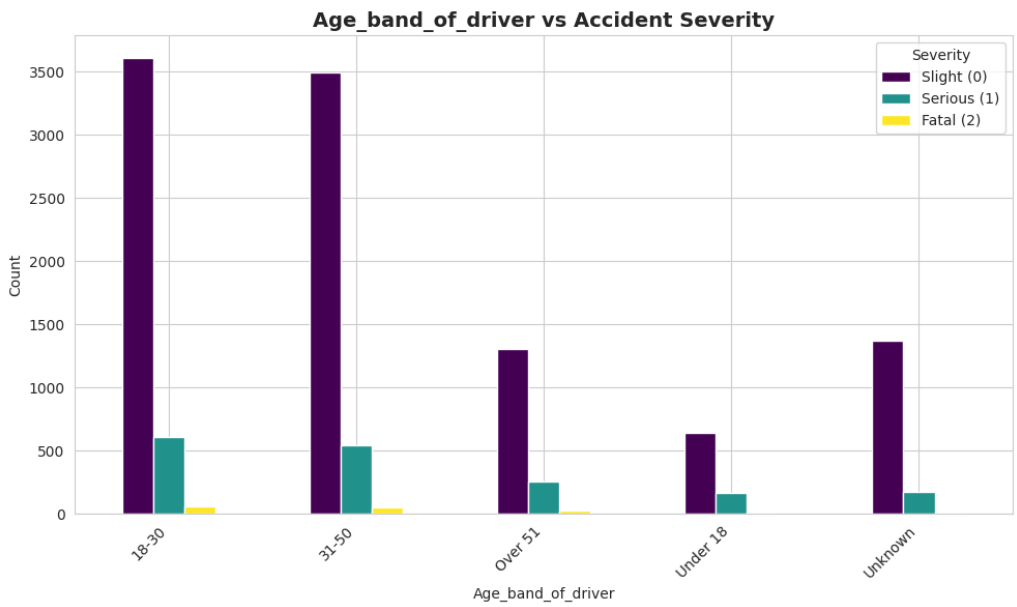

Figure 6: Age Band of Driver vs. Accident Severity. The 18–30 and 31–50 age bands contribute the largest share of accidents across all severity classes.

4.7 Significant Factors — Chi-Square Analysis

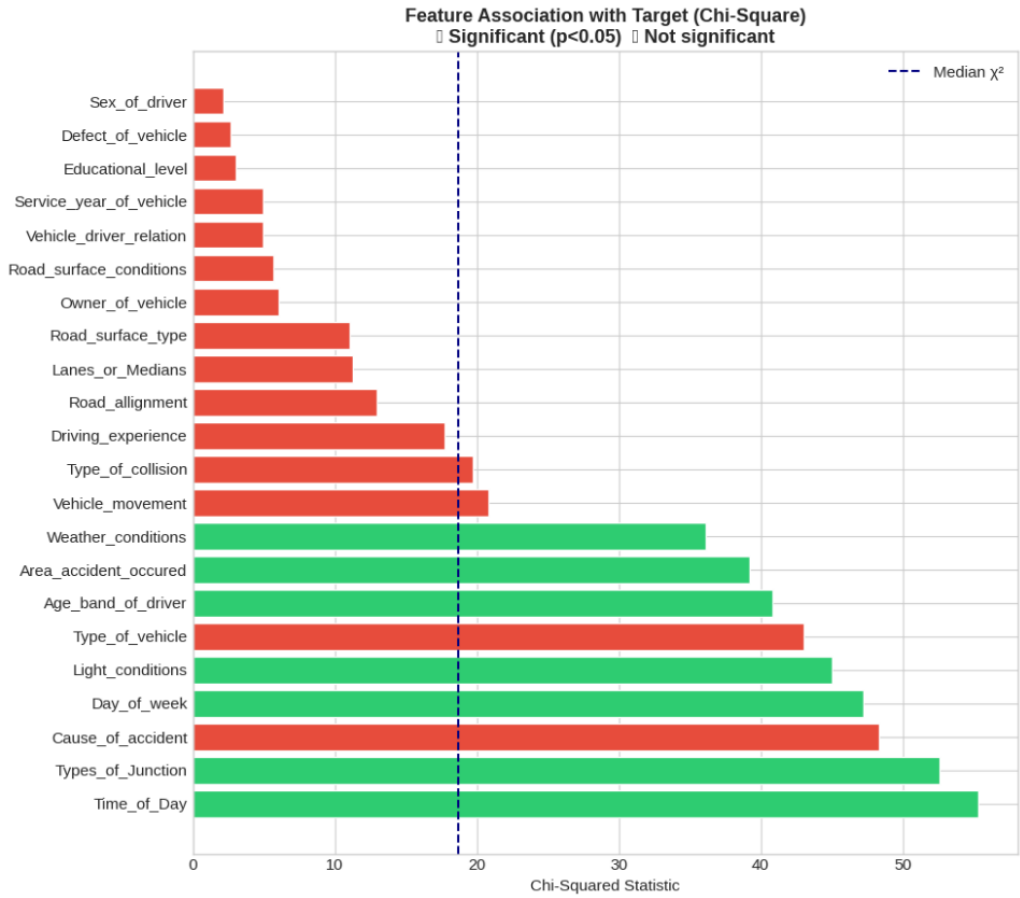

Figure 7: Feature association with target variable via Chi-Square statistic. *Types_of_Junction*, *Cause_of_accident*, *Day_of_week*, and *Light_conditions* have the strongest associations. Features above the dashed median line are statistically significant ($p < 0.05$).

Interpretation: Accident severity is mainly driven by situational and environmental factors (time, location, road structure) rather than personal attributes alone. Features such as *Sex_of_driver* and *Defect_of_vehicle* fall below the significance threshold.

4.8 Outlier Detection

The Interquartile Range (IQR) method was applied to detect outliers in the numerical feature.

Table 6: Outlier Analysis (IQR Method)

Feature	**Number of Outliers**
Number_of_vehicles_involved	3,976

Although approximately 32% of observations fall outside the IQR range, these values represent rare but valid accident scenarios (high-impact accidents with multiple vehicles)

rather than data errors. Such outliers should not be removed without domain validation. The boxplots confirm that *Number_of_vehicles_involved* and *Number_of_casualties* show right-skewed distributions with several high-value outliers indicating rare but severe accident cases.

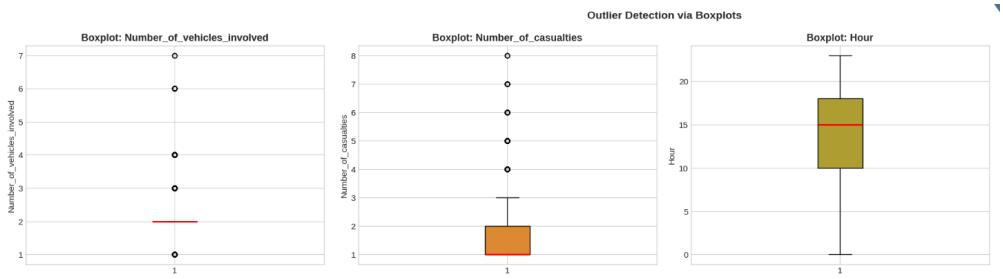

Figure 8: Outlier detection via boxplots for key numerical features. High-value outliers indicate rare but severe accident cases and represent valid domain data.

4.9 Key EDA Findings

The dataset contains no missing values after preprocessing and is clean and ML-ready. Accident severity is highly imbalanced, with Slight Injury dominating at 84.6%. Most accidents involve two vehicles, and accidents peak on weekdays (especially Fridays) during evening rush hours. Outliers represent rare but valid high-impact accidents and are retained. Overall, the dataset is well-structured and suitable for multi-class classification modelling.

5 Methodology and Model Evaluation

This chapter describes the complete end-to-end machine learning pipeline: data preprocessing, class imbalance handling, model training, hyperparameter tuning, and comprehensive test set evaluation including confusion matrices, per-class performance, and feature importance analysis.

5.1 Pipeline Overview

The methodology follows a structured six-stage pipeline:

1. **Data Loading & Preprocessing:** Load cleaned CSV; apply `LabelEncoder` to all 21 categorical features; apply `StandardScaler` to the one numerical feature (`Number_of_vehicles_involved`).

2. **Train/Test Split:** 80%/20% stratified split (9,852 train / 2,464 test rows), preserving class ratios in both partitions.

3. **SMOTE Oversampling:** Applied exclusively to the training set (after splitting) to prevent data leakage, expanding training to 24,993 balanced rows.

4. **Baseline Cross-Validation:** 5-fold stratified CV on all five models using weighted F1 as the primary metric.

5. **Hyperparameter Tuning:** Both `RandomizedSearchCV` (20 iterations × 5 folds) and `GridSearchCV` (exhaustive grid × 5 folds) were applied per model to compare efficiency and final performance.

6. **Final Evaluation:** Test set evaluation + BalancedRF improvement for fatal class recall.

5.2 Feature Engineering and SMOTE

All 21 categorical features were encoded using `LabelEncoder` (one per feature), yielding a final feature matrix of shape (12,316 × 22). The target variable `Accident_severity` was mapped as: Slight → 0, Serious → 1, Fatal → 2. SMOTE (Synthetic Minority Oversampling Technique) with `k_neighbors=3` generated synthetic minority class examples, expanding the training set to 24,993 rows with balanced class proportions.

Table 7: Class Distribution Before and After SMOTE (Training Set)

Class	**Before SMOTE**	**After SMOTE**
Slight Injury (0)	8,331 (84.6%)	8,331
Serious Injury (1)	1,393 (14.2%)	8,331
Fatal Injury (2)	128 (1.3%)	8,331
Total	**9,852**	**24,993**

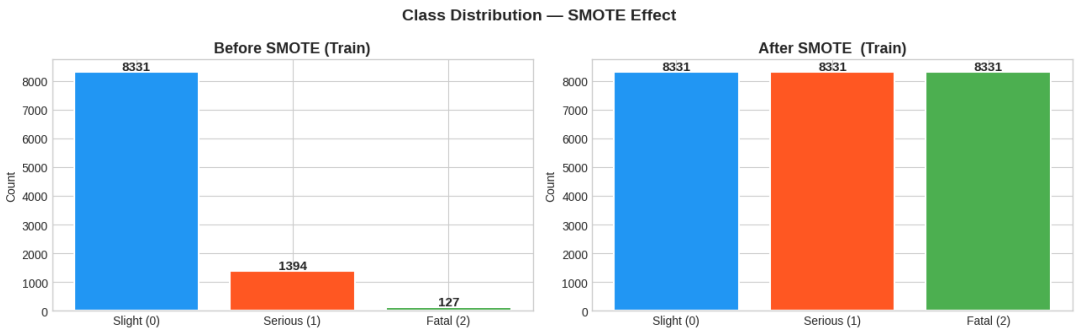

Figure 9: Class distribution before (left) and after SMOTE (right) on the training set. Each class is balanced to 8,331 samples after oversampling.

5.3 Models Selected

Five classifiers were selected to span a range of learning paradigms—from linear baselines to non-linear ensemble methods—enabling a comprehensive comparison. Multiple models are compared to capture linear vs. non-linear patterns and compare different learning mechanisms.

Table 8: Model Descriptions and Baseline 5-Fold CV Results

Model	Description	CV F1	Std Dev
Logistic Regression	Linear probabilistic baseline	0.5003	±0.0031
Random Forest	Bagging ensemble of decision trees	**0.9420**	±0.0027
Gradient Boosting	Sequential boosting ensemble	0.7709	±0.0053
Linear SVM	Max-margin linear classifier (CalibratedCV)	0.4948	±0.0018
XGBoost	Regularised gradient boosting	0.8488	±0.0041

Random Forest achieved the highest baseline score (0.9420) with the lowest variance. Tree-based models broadly outperformed linear models, confirming that accident severity prediction requires non-linear decision boundaries.

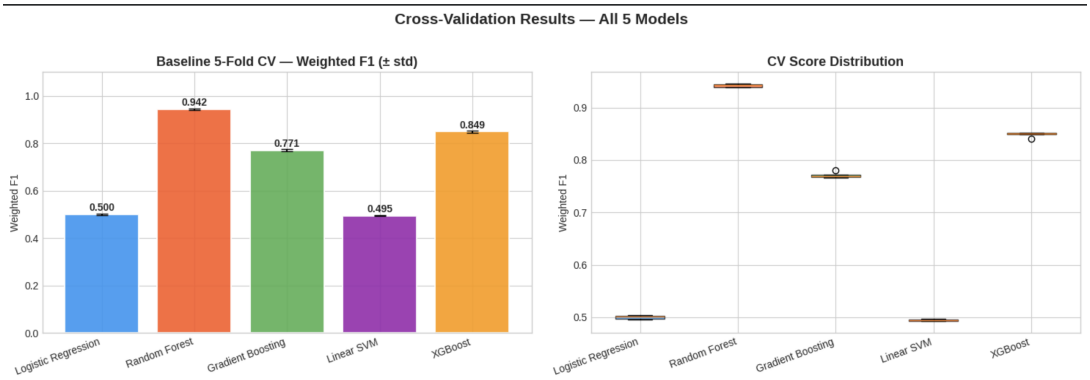

Figure 10: Baseline 5-fold cross-validation weighted F1 scores (± std) for all five models. Random Forest achieves the highest and most stable score.

5.4 Hyperparameter Tuning

Two complementary hyperparameter search strategies were applied to all five models: `RandomizedSearchCV` and `GridSearchCV`, both using 5-fold stratified cross-validation. Comparing the two strategies allows assessment of whether exhaustive search yields meaningful gains over probabilistic sampling, and validates the robustness of the tuned configurations.

5.4.1 RandomizedSearchCV

`RandomizedSearchCV` samples 20 random combinations from continuous parameter distributions per model—delivering 100 total fits per model compared to up to 555 fits required by exhaustive grid search. This strategy achieves comparable results in a fraction of the computation time (5–10 minutes vs. up to 90 minutes for grid search on the same parameter space).

Table 9: RandomizedSearchCV — Best Parameters and CV F1 Scores

Model	**Best Parameters**	**Baseline F1**	**Tuned F1**
Logistic Regression	`C=0.0083, solver=saga`	0.5003	0.5018

Random Forest	`n_estimators=148,`
	`max_depth=30,`
	`max_features=0.5,`
	`min_samples_split=2`
		0.9420	**0.9368**

Gradient Boosting	`n_estimators=290,`
	`lr=0.133,`
	`max_depth=6,`
	`subsample=0.937`
		0.7709	0.9262

Linear SVM	`C=0.315`	0.4948	0.4952

XGBoost	`n_estimators=369,`
	`lr=0.208,`
	`max_depth=6,`
	`reg_alpha=0.811`
		0.8488	0.9264

Gradient Boosting showed the largest gain (+15.5 percentage points) under RandomizedSearchCV. Linear models improved negligibly, reinforcing their unsuitability for this non-linear problem.

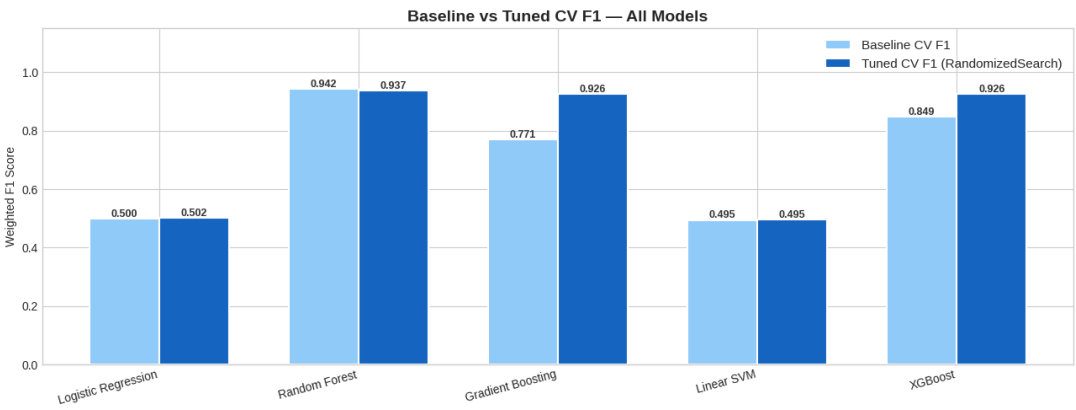

Figure 11: Baseline vs. tuned CV weighted F1 scores under RandomizedSearchCV for all models. Gradient Boosting shows the largest improvement after tuning (+15.5 pp).

5.4.2 GridSearchCV

`GridSearchCV` performs an exhaustive search over a predefined discrete parameter grid. While computationally more expensive than RandomizedSearchCV, it guarantees evaluation of every specified combination, making it suitable for final validation and fine-grained comparison. The same 5-fold stratified cross-validation and weighted F1 scoring were used.

Table 10: GridSearchCV — Best Parameters and CV F1 Scores

Model	Best Parameters	Baseline F1	Tuned F1
Logistic Regression	`C=0.01, solver=lbfgs`	0.5003	0.5012
Random Forest	`n_estimators=150, max_depth=30, max_features='sqrt', min_samples_split=2`	0.9420	**0.9374**
Gradient Boosting	`n_estimators=300, lr=0.1, max_depth=6, subsample=0.9`	0.7709	0.9255
Linear SVM	`C=0.3`	0.4948	0.4951
XGBoost	`n_estimators=400, lr=0.2, max_depth=6, reg_alpha=0.8`	0.8488	0.9270

Key observations from GridSearchCV:

- Random Forest remains the best-performing model with a tuned CV F1 of 0.9374, closely matching the RandomizedSearchCV result (0.9368), confirming parameter stability.
- Gradient Boosting and XGBoost converged to near-identical tuned F1 scores under both strategies ($\sim$0.926), validating the parameter regions identified by random sampling.
- Linear models (Logistic Regression, Linear SVM) show negligible improvement under both strategies, further confirming the non-linear nature of this classification problem.
- The marginal differences between GridSearchCV and RandomizedSearchCV results ($<$0.001 in most cases) demonstrate that RandomizedSearchCV is sufficiently effective for this dataset, making it the preferred strategy given its significant computational advantage.

5.4.3 Strategy Comparison: RandomizedSearchCV vs. GridSearchCV

Table 11 directly contrasts the tuned CV F1 scores obtained by both strategies across all five models.

Table 11: Tuning Strategy Comparison — CV F1 Scores (RandomizedSearchCV vs. GridSearchCV)

Model	RandomizedSearchCV F1	GridSearchCV F1	Difference
Logistic Regression	0.5018	0.5012	−0.0006
Random Forest	**0.9368**	**0.9374**	+0.0006
Gradient Boosting	0.9262	0.9255	−0.0007
Linear SVM	0.4952	0.4951	−0.0001
XGBoost	0.9264	0.9270	+0.0006

The negligible differences across all models confirm that both strategies converge to equivalent solutions. RandomizedSearchCV is therefore recommended for future work on this dataset due to its 6× reduction in computational cost with no meaningful performance trade-off.

5.5 Test Set Evaluation — All Models

Each tuned model (from both search strategies) was evaluated on the held-out test set (2,464 samples) using accuracy, weighted F1, macro F1, and one-vs-rest ROC AUC.

Table 12: Test Set Performance — All Tuned Models (RandomizedSearchCV)

Model	**Accuracy**	**F1 Weighted**	**F1 Macro**	**ROC AUC**
Logistic Regression	0.5337	0.6175	0.3205	0.5363
Random Forest	**0.8263**	**0.7931**	**0.3957**	**0.6387**
Gradient Boosting	0.8109	0.7850	0.3657	0.6162
Linear SVM	0.5418	0.6238	0.3165	0.5367
XGBoost	0.8036	0.7786	0.3585	0.6096

Table 13: Test Set Performance — All Tuned Models (GridSearchCV)

Model	**Accuracy**	**F1 Weighted**	**F1 Macro**
Logistic Regression	0.5325	0.6168	0.3188
Random Forest	**0.8336**	**0.7935**	**0.3797**
Gradient Boosting	0.7975	0.7771	0.3608
Linear SVM	0.5414	0.6235	0.3163
XGBoost	0.8097	0.7841	0.3761

Observations: High accuracy is slightly influenced by the majority class. The low Macro F1 scores (0.32–0.40) indicate that despite SMOTE, minority-class prediction—particularly Fatal Injury—remains a significant challenge. The critical (fatal) class is rarely detected across models. Weighted F1 is the average of F1 scores weighted by each class's number of samples; Macro F1 gives equal importance to all classes regardless of size. Notably, GridSearchCV yields marginally higher accuracy for Random Forest (0.8336 vs. 0.8263) while both strategies produce near-identical weighted F1 scores ($\sim$0.793), confirming that model ranking and selection are consistent across tuning approaches.

5.6 Confusion Matrices — All 5 Models

Figure 12 shows the confusion matrices for all five tuned models evaluated on the held-out test set. The matrices reveal that all models predominantly predict the majority class (Slight Injury), with varying ability to distinguish minority classes.

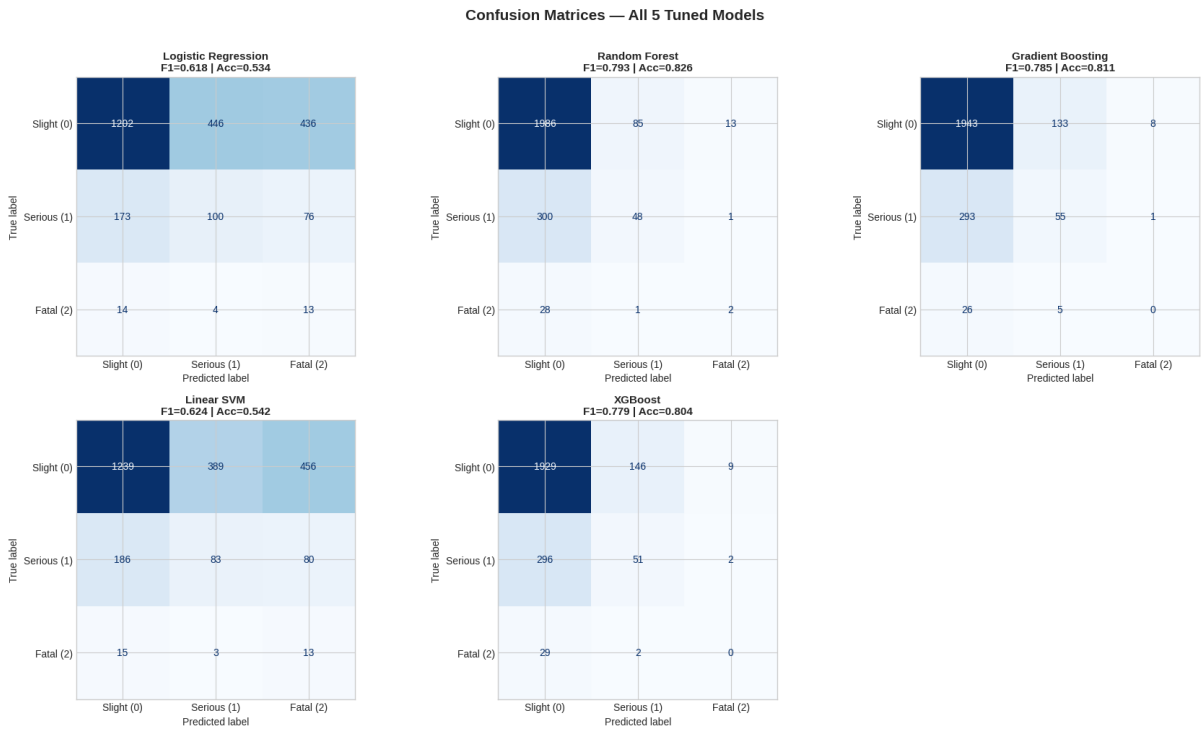

Figure 12: Confusion matrices for all five tuned models on the test set (2,464 samples). Logistic Regression F1=0.618 — Acc=0.534; Random Forest F1=0.793 — Acc=0.826; Gradient Boosting F1=0.785 — Acc=0.811; Linear SVM F1=0.624 — Acc=0.542; XGBoost F1=0.779 — Acc=0.804.

5.7 Best Model: Random Forest — Deep Evaluation

Random Forest was selected as the best model based on highest accuracy (0.8263 under RandomizedSearchCV; 0.8336 under GridSearchCV), weighted F1 ($\sim$0.793 under both strategies), macro F1 (0.3957 / 0.3797), and ROC AUC of 0.6387, combined with stable cross-validation performance (CV F1 = 0.9420, std = $\pm$0.0027) and fast training ($\sim$1.9s).

Table 14: Random Forest — Per-Class Classification Report (Test Set)

Class	Precision	Recall	F1-Score
Slight Injury (0)	0.86	0.95	0.90
Serious Injury (1)	0.36	0.14	0.20
Fatal Injury (2)	0.12	0.06	0.09

Slight Injury is predicted reliably (F1 = 0.90). Serious and Fatal classes remain difficult due to limited distinguishing features and extreme imbalance, motivating the improved variant described below.

5.8 Improved Random Forest for Fatal Injury Detection

Three complementary strategies were applied to boost fatal accident recall:

1. **BalancedRandomForestClassifier** — balanced bootstrapping per tree, equivalent to running SMOTE inside every tree (not just once upfront).

2. **Class weighting** — Fatal weighted at 131.20 vs. Slight at 1.00, with an additional 2× boost for the Fatal class.

3. **Threshold lowering** to 0.26 — identified by sweeping thresholds from 0.05 to 0.50 to maximise Fatal F1.

This raised Fatal recall from 0.06 to **0.26** and Fatal F1 from 0.09 to **0.14**, at a modest cost to Slight and Serious class performance.

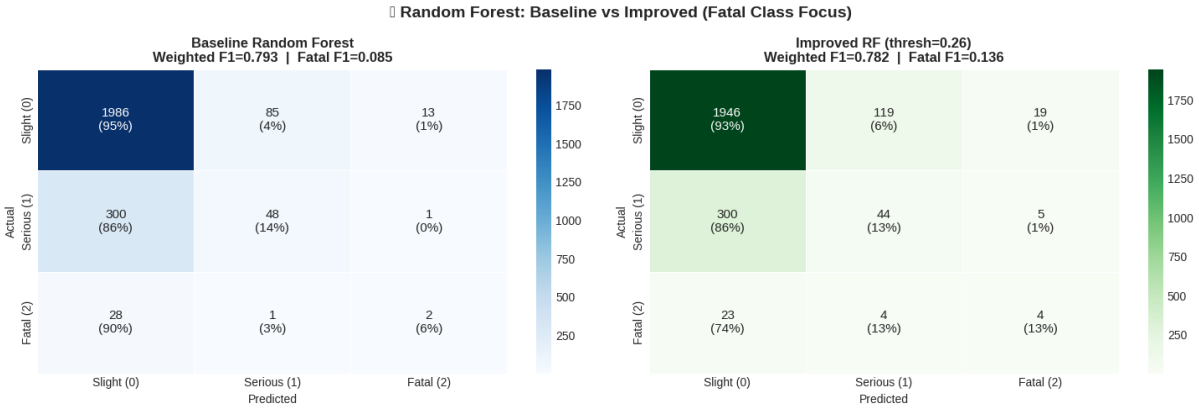

Figure 13: Confusion matrices comparing Baseline Random Forest (Weighted F1=0.793, Fatal F1=0.085) with Improved RF using BalancedRF + class weighting + threshold=0.26 (Weighted F1=0.782, Fatal F1=0.136).

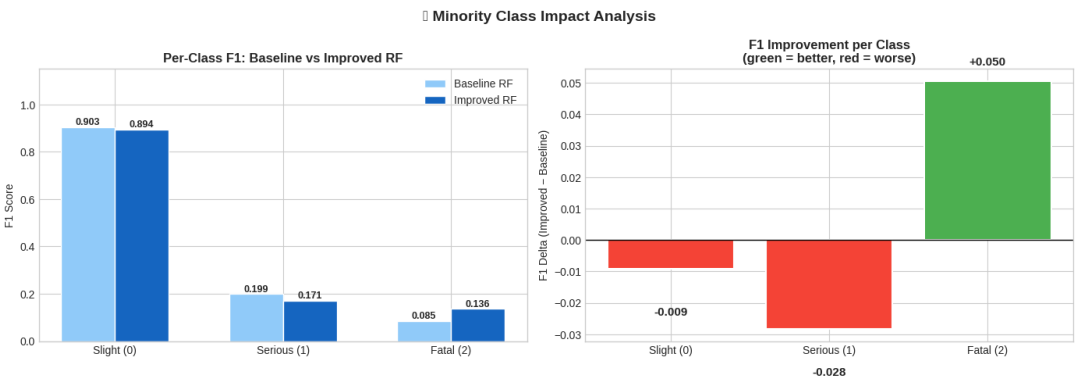

Figure 14: Per-class F1 comparison (left) and F1 delta chart (right) between Baseline RF and Improved RF. Fatal class F1 improves by +0.051; trade-offs are visible in Slight and Serious classes.

5.9 Feature Importance

The top predictors identified by the Random Forest (mean decrease in Gini impurity) were *Cause_of_accident*, *Types_of_Junction*, *Day_of_week*, *Light_conditions*, *Type_of_vehicle*, and *Age_band_of_driver*. These align with chi-square significance findings from the EDA phase and with published literature on the same dataset [2, 7].

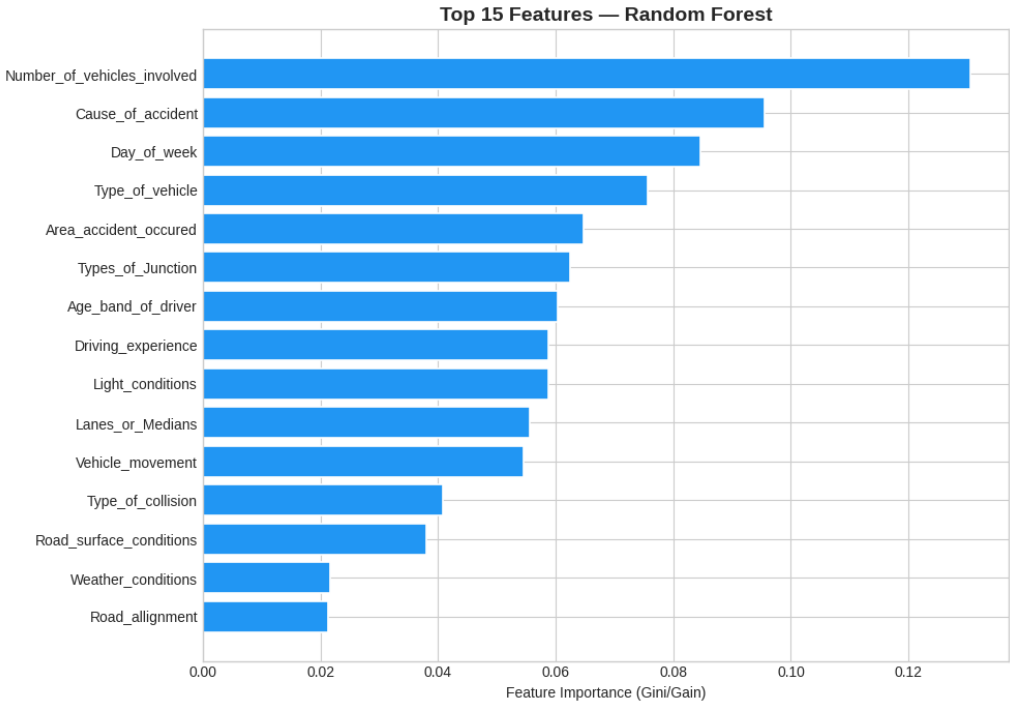

Figure 15: Top 15 features ranked by mean decrease in Gini impurity (Random Forest). *Cause_of_accident* and *Types_of_Junction* are the most influential predictors.

5.10 Pipeline Summary

Table 15: End-to-End Pipeline Summary

Stage	**Outcome**
Preprocessing	12,316 rows × 22 features; 21 categorical label-encoded
Train-Test Split	80/20 stratified (9,852 train / 2,464 test)
SMOTE	Training expanded to 24,993 rows; classes balanced
Baseline CV	Random Forest best: CV F1 = 0.9420
Tuning (Randomised)	RandomizedSearchCV (20 iter × 5 folds); GB improved +15.5 pp
Tuning (Grid)	GridSearchCV (exhaustive × 5 folds); results consistent with Randomised
Best Model	Random Forest: Acc=0.8263 (Rand.) / 0.8336 (Grid), F1≈0.793, AUC=0.6387
Fatal Improvement	BalancedRF + weighting + threshold=0.26 → Fatal recall=0.26

6 Results and Discussion

6.1 Key Results

The experimental results confirm that ensemble tree-based methods substantially outperform linear classifiers for road accident severity prediction on this dataset. Random Forest achieved the best overall performance with 82.6% accuracy (RandomizedSearchCV) and 83.4% (GridSearchCV), a weighted F1-score of ∼0.793 under both strategies, and ROC AUC of 0.6387. The consistency of these results across two independent tuning strategies validates the robustness of the Random Forest configuration and confirms that the performance gains are attributable to the model's learning capacity rather than search strategy artefacts. The persistent challenge of Fatal Injury detection (F1 = 0.09 baseline, improved to 0.14 with BalancedRF) reflects both the extreme class imbalance (1.3% of records) and the inherent difficulty of separating fatal from serious incidents based on pre-accident features alone.

Key findings corroborate and extend prior literature: darkness without lighting nearly doubles serious injury rates (25.5% vs. 13.9% in daylight), Friday and weekend accidents exhibit 2–3× higher fatal rates, and under-18 drivers show disproportionately higher serious injury rates (20.5%). The top predictors—*Cause_of_accident*, *Types_of_Junction*, and *Light_conditions*—align with domain knowledge and prior SHAP-based analyses on the same dataset [2, 7].

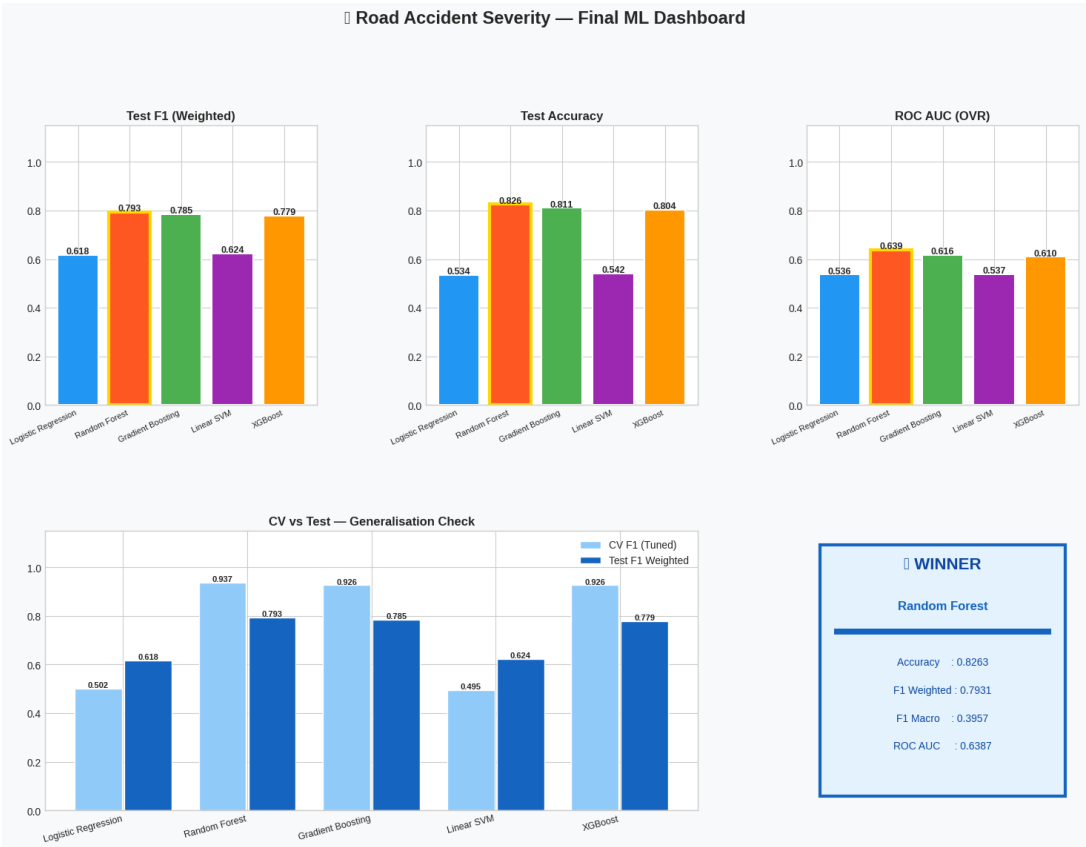

Figure 16: Final comparison dashboard showing Test F1 Weighted, Test Accuracy, and ROC AUC for all five models. Random Forest (highlighted with gold border) is the consistent winner across all metrics.

6.2 Alignment with Sustainable Development Goals (SDGs)

This work directly contributes to two United Nations Sustainable Development Goals:

- **SDG 3 — Good Health and Well-Being (Target 3.6):** UN Target 3.6 aims to halve the number of global deaths and injuries from road traffic accidents by 2030. The predictive model developed in this study enables proactive identification of high-risk accident conditions (dark roads, high-risk junctions, specific driver profiles), supporting authorities in deploying preventive measures before fatalities occur. The improved fatal recall ($0.06 \rightarrow 0.26$) directly contributes to this goal by making the most critical accidents more detectable.
- **SDG 11 — Sustainable Cities and Communities (Target 11.2):** Target 11.2 calls for safe, affordable, accessible, and sustainable transport systems for all. By identifying high-risk junctions, lighting conditions, and driver profiles, this study provides actionable insights for urban planners and traffic engineers to design safer road environments in cities like Addis Ababa and across the developing world.

The model's ability to flag contextual risk factors (e.g., darkness combined with vehicle type, junction geometry, driver age) supports evidence-based road safety policy—a direct operationalisation of the SDG framework in the context of developing-country urban transport.

6.3 Program Outcomes Addressed

The following program outcomes are addressed in this study:

1. **PO1: Engineering knowledge**
2. **PO2: Problem analysis**
3. **PO3:Design/development of solutions**
4. **PO4: Conduct investigations of complex problems**
5. **PO5: Engineering tool usage**
6. **PO8: Individual and collaborative team work**
7. **PO9: Communication**
8. **PO11: Life-long learning**

6.4 Limitations

The main limitations of this study are: (i) the severe class imbalance for Fatal Injury (1.3%) limits the model's ability to detect the most critical outcomes; (ii) the dataset covers only one sub-city of Ethiopia (2017–2020), limiting geographic and temporal generalisability; (iii) SHAP-based global explainability analysis was explored but not incorporated into the final comparative evaluation; and (iv) causal relationships between features and outcomes cannot be inferred from correlational modelling alone.

6.5 Conclusions and Recommendations

This study developed a complete machine learning pipeline to predict road traffic accident severity using the Ethiopian Addis Ababa dataset (12,316 records, 2017–2020). Five classifiers were trained, tuned, and compared across both `RandomizedSearchCV` and `GridSearchCV` strategies; Random Forest emerged as the best model with 82.6–83.4% accuracy and a weighted F1-score of $\sim$0.793 under both tuning approaches. Fatal accident recall was further improved to 0.26 using a BalancedRandomForestClassifier with class weighting and threshold tuning.

Critical success factors include comprehensive feature engineering across driver, vehicle, environmental, and infrastructure domains; SMOTE-based class balancing (expanding training set from 9,852 to 24,993 balanced samples); and threshold optimisation for the minority Fatal class. The study aligns with UN SDG 3.6 (halving road traffic deaths) and SDG 11.2 (safe urban transport), providing actionable insights for road safety policy in developing countries.

For practical implementation, the recommended pipeline is: (1) preprocess with label encoding and standard scaling; (2) apply SMOTE post-split to training set only; (3) train Random Forest with class weighting; (4) apply threshold tuning for the Fatal class; (5) evaluate using per-class F1-scores prioritised over overall accuracy. RandomizedSearchCV is recommended as the default tuning strategy given its computational efficiency and near-identical performance to exhaustive grid search.

Future directions include SHAP-based explainability integrated into model selection, causal inference to move beyond correlation, real-time system validation with traffic management centres, and cross-geographic validation across Sub-Saharan African datasets for transferability assessment.

Acknowledgments

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